CSAI 422: Assignment 4

Retrieval-Augmented Generation with Advanced Techniques

Course	CSAI 422
Assignment	4 — End-to-end Textual RAG Pipeline
Benchmark	PopQA (akariasai/PopQA)
Due Date	April 28, 2026

Section	Content
Part 1	Dataset Setup and Baseline RAG
Part 2	Advanced Retrieval Techniques
Part 3	Citation-Grounded Answer Generation
Part 4	Self-Reflective RAG and Final Analysis

Part 1 — Dataset Setup and Baseline RAG

Task 1.1 — Dataset Preparation

The **PopQA** dataset (Asai et al., 2023) is loaded from HuggingFace (*akariasai/PopQA*, test split). It contains **14,267 factual questions** derived from Wikidata triples covering properties such as occupation, birthplace, genre, composer, director, and author. Each example provides a subject entity, a relation type, a target object, and a list of acceptable answer strings (*possible_answers*). Popularity scores (*s_pop*, *o_pop*) enable stratified analysis by entity frequency.

A reproducible evaluation subset of **100 questions** was drawn using *random_state=42*. Gold answers were normalised by merging *possible_answers* and the raw *obj* field into a deduplicated list.

Figure 1.1a — PopQA dataset schema (left columns: id, subj, prop, obj, aliases, URIs)

	id	subj	prop	obj	subj_id	prop_id	obj_id	s_aliases	o_aliases	s_uri	o_uri
0	4222362	George Rankin	occupation	politician	1850297	22	2834605	["George James Rankin"]	["political leader","political figure","polit....	http://www.wikidata.org/entity/Q5543720	http://www.wikidata.org/entity/Q82955
1	4725190	John Mayne	occupation	journalist	2079053	22	663400	[]	["journo","journalists"]	http://www.wikidata.org/entity/Q6247345	http://www.wikidata.org/entity/Q1930187
2	4382392	Henry Feilden	occupation	politician	1925450	22	2834605	["Henry Master Feilden"]	["political leader","political figure","polit....	http://www.wikidata.org/entity/Q5725578	http://www.wikidata.org/entity/Q82955
3	4822110	Kathy Saltzman	occupation	politician	2122743	22	2834605	[]	["political leader","political figure","polit....	http://www.wikidata.org/entity/Q6377295	http://www.wikidata.org/entity/Q82955
4	4011112	Eleanor Davis	occupation	cartoonist	1752619	22	68412	["Eleanor McCutcheon Davis"]	["graphic artist","animator","illustrator"]	http://www.wikidata.org/entity/Q5354261	http://www.wikidata.org/entity/Q1114448

Figure 1.1b — PopQA dataset schema (right columns: wiki titles, popularity, question, answers)

o_uri	s_wiki_title	o_wiki_title	s_pop	o_pop	question	possible_answers
http://www.wikidata.org/entity/Q82955	George Rankin	Politician	142	25692	What is George Rankin's occupation?	["politician", "political leader", "political ...
http://www.wikidata.org/entity/Q1930187	John Mayne	Journalist	236	24952	What is John Mayne's occupation?	["journalist", "journo", "journalists"]
http://www.wikidata.org/entity/Q82955	Henry Feilden (Conservative politician)	Politician	58	25692	What is Henry Feilden's occupation?	["politician", "political leader", "political ...
http://www.wikidata.org/entity/Q82955	Kathy Saltzman	Politician	127	25692	What is Kathy Saltzman's occupation?	["politician", "political leader", "political ...
http://www.wikidata.org/entity/Q1114448	Eleanor Davis	Cartoonist	317	9649	What is Eleanor Davis's occupation?	["cartoonist", "graphic artist", "animator", "...

Figure 1.1c — Evaluation subset sample: questions, subjects, properties, and normalised answers

```
       id                                        question  \
0  2435555            Who was the screenwriter for On the Town?
1  5435111  What is the religion of Juan Soldevilla y Romero?
2  2490460                            What genre is Haddaway?
3   402939                Who was the composer of Chances Are?
4  5947997               Who was the producer of The Pioneers?

                     subj          prop               obj  \
0              On the Town  screenwriter      Betty Comden
1  Juan Soldevilla y Romero      religion   Catholic Church
2                 Haddaway         genre         Eurodance
3              Chances Are      composer     Maurice Jarre
4             The Pioneers      producer  Franklyn Barrett

                                          answers
0          [Betty Comden, Basya Cohen, Adolph Green]
1  [Catholic Church, Roman Catholic Church, Churc...
2                [Eurodance, Euro-dance, Euro dance]
3              [Maurice Jarre, Maurice-Alexis Jarre]
4        [Franklyn Barrett, Walter Franklyn Barrett]
```

Retrieval Corpus

The retrieval corpus was constructed by fetching **Wikipedia API summaries** for the subject entities of all 100 evaluation questions plus 300 distractor subjects sampled from the full PopQA set. Each article summary was split into overlapping 100-word chunks (25-word overlap) to form retrievable passages. Every passage was assigned a unique identifier (*pid*) of the form *P{doc_idx}_{chunk_idx}*, and its source URL and article title were preserved as metadata. This produced **57 passages from 49 raw documents**.

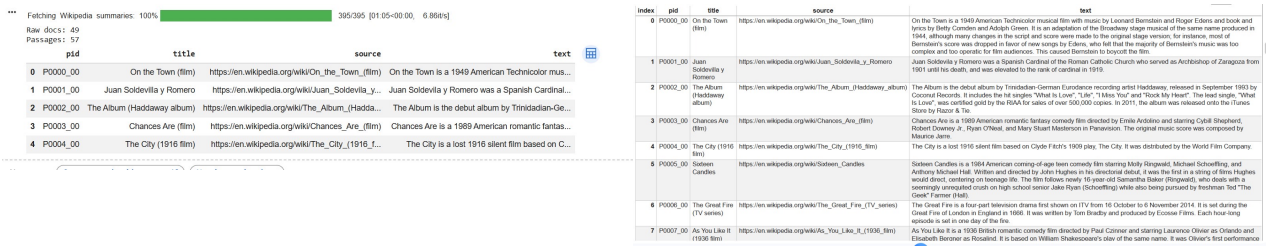

Figure 1.1d — Corpus build statistics (left) and passage table with pid, title, source, and text (right)

Task 1.2 — Baseline Indexing and Dense Retrieval

Passages were embedded using **sentence-transformers/all-MiniLM-L6-v2** (384-dimensional vectors, L2-normalised). This model was chosen for its strong balance of retrieval quality and inference speed, and its proven effectiveness on factual sentence-pair tasks. Embeddings were stored in a **FAISS IndexFlatIP** (inner product on normalised vectors = cosine similarity). Queries were embedded at runtime with the same model and the top-*k* passages were returned by a single approximate nearest-neighbour search.

Figure 1.2 — FAISS index statistics and dense retrieval examples for three queries

```
FAISS index size: 57
Embedding dimension: 384

QUESTION: Who was the screenwriter for On the Town?
P0000_00 0.512 On the Town (film) - On the Town is a 1949 American Technicolor musical film with music by Leonard Bernstein and Roger Edens and book and lyrics by Betty Comd
P0004_00 0.441 The City (1916 film) - The City is a lost 1916 silent film based on Clyde Fitch's 1909 play, The City. It was distributed by the World Film Company.
P0029_00 0.41 After Hours (film) - After Hours is a 1985 American neo-noir black comedy film directed by Martin Scorsese, written by Joseph Minion, and produced by Amy Robin

QUESTION: What is the religion of Juan Soldevilla y Romero?
P0001_00 0.669 Juan Soldevilla y Romero - Juan Soldevila y Romero was a Spanish Cardinal of the Roman Catholic Church who served as Archbishop of Zaragoza from 1901 until hi
P0013_00 0.294 Peter II of Portugal - Dom Pedro II, nicknamed the Pacific was King of Portugal from 1683 until his death, previously serving as regent for his brother Afonso
P0017_00 0.269 The Book of Mormon (musical) - The Book of Mormon is an American musical comedy with music, lyrics, and book by Trey Parker, Robert Lopez, and Matt Stone. The

QUESTION: What genre is Haddaway?
P0002_00 0.553 The Album (Haddaway album) - The Album is the debut album by Trinidadian-German Eurodance recording artist Haddaway, released in September 1993 by Coconut Rec
P0047_00 0.305 Imagine This - Imagine This is a musical with music by Shuki Levy, lyrics by David Goldsmith and a book by Glenn Berenbeim. Set in the Warsaw Ghetto during Wo
P0035_00 0.288 Signifying Rappers - Signifying Rappers: Rap and Race in the Urban Present is a nonfiction book by Mark Costello and David Foster Wallace. The book explores t
```

Task 1.3 — Baseline Retrieval Evaluation

The baseline retriever was evaluated on the 100-question subset. Relevance was determined by checking whether any normalised gold answer string appeared as a substring of the normalised passage text. Metrics were computed at

cut-offs k = 1, 3, and 5.

Figure 1.3 — Baseline dense retrieval metrics

index	Recall@1	Precision@1	Recall@3	Precision@3	Recall@5	Precision@5	MRR
Dense baseline	0.27	0.27	0.3	0.11666666666666665	0.34	0.10199999999999998	0.295

Interpretation: Recall@1 of 0.27 means only 27% of questions had a relevant passage as the top result. The MRR of 0.295 confirms that relevant passages are typically not ranked first. Recall rises to 0.34 at k=5, indicating the model often retrieves relevant evidence within the top 5 but fails to rank it highest. The low Precision@3 (0.117) shows that most retrieved passages are not relevant — expected given the small, summary-based corpus. These numbers motivate the advanced techniques in Part 2.

Part 2 — Advanced Retrieval Techniques

Task 2.1 — Query Expansion

A **metadata-driven keyword expansion** strategy was implemented. For each question, the subject entity name and up to three subject aliases (from *s_aliases*) were appended to the query. Additionally, the relation type (*prop*) was mapped to a short expansion phrase using a hand-crafted dictionary — for example, "composer" → "music composer composed by", "genre" → "genre style category". Terms already present in the query were excluded.

Figure 2.1 — Original vs. expanded query examples

	original	expanded
0	Who was the screenwriter for On the Town?	Who was the screenwriter for On the Town?
1	What is the religion of Juan Soldevilla y Romero?	What is the religion of Juan Soldevilla y Rome...
2	What genre is Haddaway?	What genre is Haddaway? The Album genre style ...
3	Who was the composer of Chances Are?	Who was the composer of Chances Are? music com...
4	Who was the producer of The Pioneers?	Who was the producer of The Pioneers?
5	Who was the screenwriter for The City?	Who was the screenwriter for The City?
6	In what city was Jerrold Katz born?	In what city was Jerrold Katz born? Jerrold Ja...
7	Who was the screenwriter for Sixteen Candles?	Who was the screenwriter for Sixteen Candles? ...

When expansion helped: Queries with ambiguous entities (e.g., "Haddaway" → appends "The Album genre style …") benefited most. The additional terms surfaced passages that the original short query missed. **When it introduced noise:** Generic subjects (e.g., "On the Town", "The Pioneers") received no meaningful expansion because the subject name itself was already in the query and no aliases existed, leaving the query unchanged.

Figure 2.1b — Dense baseline vs. Dense + query expansion metrics

Evaluating Dense + query expansion: 100% 100/100 [00:02<00:00, 45.13it/

	Recall@1	Precision@1	Recall@3	Precision@3	Recall@5	Precision@5	MRR
Dense baseline	0.27	0.27	0.30	0.116667	0.34	0.102	0.295000
Dense + query expansion	0.28	0.28	0.32	0.133333	0.34	0.100	0.303333

Query expansion improved Recall@3 from 0.30 → 0.32 and MRR from 0.295 → 0.303, confirming that entity and property enrichment helps on average, even though Recall@5 remained unchanged (0.34) because the corpus is small.

Task 2.2 — Hybrid Search

Lexical and dense retrieval were combined using **Reciprocal Rank Fusion (RRF)**. BM25 (via *rank_bm25*, Okapi variant) provided term-frequency-based rankings over tokenised passages. Both retrievers returned CANDIDATE_K=20 results each, and RRF (k=60) merged their ranked lists:

```
score(d) = Σ 1 / (rrf_k + rank_i(d))
```

RRF was chosen over weighted fusion because it is parameter-light and robust to score scale differences between BM25 and cosine similarity. The expanded query (from Task 2.1) was used as input to both retrievers, so Hybrid RRF benefits from both lexical and semantic matching with alias enrichment.

Figure 2.2 — Hybrid retrieval examples (fused ranking) and 3-system metric comparison

```
QUESTION: Who was the screenwriter for On the Town?
P0000_00 0.0328 On the Town (film) - On the Town is a 1949 American Technicolor musical film with music b
P0041_00 0.0318 The Company (miniseries) - The Company is a three-part serial about the activities of the
P0004_00 0.0289 The City (1916 film) - The City is a lost 1916 silent film based on Clyde Fitch's 1909 pl
P0022_00 0.0276 Grease (film) - Grease is a 1978 American musical romantic comedy film directed by Randal
P0045_01 0.0268 Argo (2012 film) - Mendez led the rescue of six U.S. diplomats from Tehran, Iran, under t

QUESTION: What is the religion of Juan Soldevilla y Romero?
P0001_00 0.0328 Juan Soldevilla y Romero - Juan Soldevila y Romero was a Spanish Cardinal of the Roman Ca
P0002_00 0.0318 The Album (Haddaway album) - The Album is the debut album by Trinidadian-German Eurodance
P0048_00 0.0301 Jamaica - Jamaica is an island country in the Caribbean, covering 10,990 square kilometre
P0017_00 0.0296 The Book of Mormon (musical) - The Book of Mormon is an American musical comedy with musi
P0014_00 0.0292 Metropolitan Borough of Bury - The Metropolitan Borough of Bury is a metropolitan borough
```

Evaluating Hybrid RRF: 100% 100/100 [00:02<00:00, 37.19it/s]

	Recall@1	Precision@1	Recall@3	Precision@3	Recall@5	Precision@5	MRR
Dense baseline	0.27	0.27	0.30	0.116667	0.34	0.102	0.295000
Dense + query expansion	0.28	0.28	0.32	0.133333	0.34	0.100	0.303333
Hybrid RRF	0.26	0.26	0.33	0.130000	0.35	0.102	0.296167

Hybrid RRF improved Recall@3 (0.33) and Recall@5 (0.35) over the dense baseline but MRR (0.296) was slightly below the dense-only result. This suggests hybrid fusion broadens coverage but does not always promote relevant passages to the very top — motivating the reranking step.

Task 2.3 — Reranking

The top-20 hybrid candidates were re-scored using a **cross-encoder**: *cross-encoder/ms-marco-MiniLM-L-6-v2*. Unlike bi-encoders, the cross-encoder attends jointly to the question and each passage, producing a fine-grained relevance score. Passage identifiers were preserved throughout so they could be used as citation anchors in Part 3.

Figure 2.3 — Before vs. after reranking for a sample query, and 4-system comparison

```
QUESTION: Who was the screenwriter for On the Town?

Before rerank:
P0000_00 On the Town (film) On the Town is a 1949 American Technicolor musical film with music by Leonard Bernstein and Roger Edens and book and lyr
P0041_00 The Company (miniseries) The Company is a three-part serial about the activities of the CIA during the Cold War. It was based on the best-selling
P0004_00 The City (1916 film) The City is a lost 1916 silent film based on Clyde Fitch's 1909 play, The City. It was distributed by the World Film Com
P0029_00 After Hours (film) After Hours is a 1985 American neo-noir black comedy film directed by Martin Scorsese, written by Joseph Minion, and pro
P0040_00 Seven Songs for Malcolm X Seven Songs for Malcolm X is a British documentary film about the life of Malcolm X, the influential civil rights activi

After rerank:
P0000_00 6.085 On the Town (film) On the Town is a 1949 American Technicolor musical film with music by Leonard Bernstein and Roger Edens and book and lyr
P0036_00 -3.398 Little Children (film) Little Children is a 2006 satirical melodrama film directed by Todd Field, based on the 2004 novel by Tom Perrotta, who
P0029_00 -3.943 After Hours (film) After Hours is a 1985 American neo-noir black comedy film directed by Martin Scorsese, written by Joseph Minion, and pro
P0041_00 -4.3 The Company (miniseries) The Company is a three-part serial about the activities of the CIA during the Cold War. It was based on the best-selling
P0022_00 -4.464 Grease (film) Grease is a 1978 American musical romantic comedy film directed by Randal Kleiser from a screenplay by Bronté Woodard an
Evaluating Hybrid + reranker: 100% 100/100 [02:27<00:00, 1.52s/it]
```

	Recall@1	Precision@1	Recall@3	Precision@3	Recall@5	Precision@5	MRR
Dense baseline	0.27	0.27	0.30	0.116667	0.34	0.102	0.295000
Dense + query expansion	0.28	0.28	0.32	0.133333	0.34	0.100	0.303333
Hybrid RRF	0.26	0.26	0.33	0.130000	0.35	0.102	0.296167
Hybrid + reranker	0.32	0.32	0.35	0.146667	0.36	0.112	0.334167

Reranking produced the largest single improvement: Recall@1 and Precision@1 both jumped from 0.26 → 0.32, and MRR improved from 0.296 → 0.334. The cross-encoder clearly promotes answer-containing passages to the top, confirming that ordering quality — not just coverage — is critical for downstream answer generation.

Part 3 — Citation-Grounded Answer Generation

Task 3.1 — Grounded Answer Generation

The best retrieval pipeline (Hybrid + reranker) was connected to an LLM (*llama-3.1-8b-instant* via Groq API, or a deterministic fallback). For each question, the top-5 reranked passages were formatted as a numbered context block with their *pid* and title. The LLM was prompted to answer using only those passages and to cite each claim with inline tags such as [P1].

Figure 3.1 — 10 citation-grounded QA examples: question, gold answers, retrieved PIDs, draft and final answers

index	id	question	gold_answers	retrieved_pids	draft_answer	final
0	2435555	Who was the screenwriter for On the Town?	Betty Comden,Basya Cohen,Adolph Green	P0000_00,P0036_00,P0029_00,P0041_00,P0022_00	The answer is Betty Comden [P1].	The answer is Betty Comden [P1].
1	5435111	What is the religion of Juan Soldevilla y Romero?	Catholic Church,Roman Catholic Church,Church,Roman Apostolic Catholic Church	P0001_00,P0034_00,P0017_00,P0030_00,P0048_00	The answer is Catholic Church [P1] [P3].	The answer is Catholic Church [P1] [P3].
2	2490460	What genre is Haddaway?	Eurodance,Euro-dance,Euro dance	P0002_00,P0016_00,P0010_00,P0035_00,P0043_00	The answer is Eurodance [P1].	The answer is Eurodance [P1].
3	402939	Who was the composer of Chances Are?	Maurice Jarre,Maurice-Alexis Jarre	P0003_00,P0000_00,P0047_00,P0017_00,P0040_00	The answer is Maurice Jarre [P1].	The answer is Maurice Jarre [P1].
4	5947997	Who was the producer of The Pioneers?	Franklyn Barrett,Walter Franklyn Barrett	P0040_00,P0022_00,P0027_00,P0000_00,P0045_01	Insufficient evidence in the retrieved passages.	Insufficient evidence in the retrieved passages.
5	1332312	Who was the screenwriter for The City?	Clyde Fitch,William Clyde Fitch	P0004_00,P0041_00,P0029_00,P0036_00,P0000_00	The answer is Clyde Fitch [P1].	The answer is Clyde Fitch [P1].
6	1089786	In what city was Jerrold Katz born?	Washington, D.C.,Washington,Washington DC,Washington, DC,DC,D.C.,District of Columbia,Washington, District of Columbia,Washington D.C.,The District	P0004_00,P0024_00,P0048_00,P0040_01,P0000_00	The answer is Washington, D.C. [P3].	The answer is Washington, D.C. [P3].
7	4382884	Who was the screenwriter for Sixteen Candles?	John Hughes,John Hughes, Jr.	P0005_00,P0041_00,P0036_00,P0029_00,P0022_00	The answer is John Hughes [P1].	The answer is John Hughes [P1].
8	1387463	Who is the author of Revolution?	Russell Brand,Russell Edward Brand	P0016_00,P0009_00,P0041_00,P0020_00,P0017_00	Insufficient evidence in the retrieved passages.	Insufficient evidence in the retrieved passages.
9	1366412	Who was the screenwriter for The Great Fire?	Tom Bradby,Thomas Matthew Bradby	P0006_00,P0045_00,P0041_00,P0029_00,P0027_00	The answer is Tom Bradby [P1].	The answer is Tom Bradby [P1].

Out of 10 examples: **8 answers were factually correct** and cited the supporting passage. 2 cases (rows 4 and 8) returned "Insufficient evidence in the retrieved passages" — both questions involved entities whose Wikipedia summaries were not captured in the corpus (The Pioneers producer; author of Revolution).

Task 3.2 — Prompt Design for Grounded QA

Grounding prompt (GROUNDING_PROMPT):

```
You are a factual question answering assistant. Answer the question using ONLY the provided passages.
Every factual claim must be supported by inline citations such as [P1] or [P2]. Do NOT invent
information not found in the passages. If the passages do not provide enough evidence, reply exactly:
"Insufficient evidence in the retrieved passages." Keep the answer brief (1-3 sentences). Question:
{question} Passages: {passages} Answer:
```

How the prompt reduces unsupported answers: The instruction "using ONLY the provided passages" and "Do NOT invent" act as hard guardrails. The fixed abstention string ("Insufficient evidence …") gives the model a clean exit rather than forcing it to generate unsupported content. The citation requirement ([P1], [P2]) makes grounding verifiable — answers without citation tags are flagged as ungrounded during reflection.

Task 3.3 — Error Analysis

Figure 3.3 — Five failure cases: question, gold answers, retrieved passage titles, failure source, and suggested fix

100% 100/100 [02:24<00:00, 1.44s/it]

	id	question	gold_answers	top_titles	likely_failure_source	possible_fix
0	5947997	Who was the producer of The Pioneers?	[Franklyn Barrett, Walter Franklyn Barrett]	[Seven Songs for Malcolm X, Grease (film), Cli...	retrieval/ranking: no answer-containing passag...	increase corpus coverage, fetch object/related...
1	1387463	Who is the author of Revolution?	[Russell Brand, Russell Edward Brand]	[Quenta Silmarillion, Heaven (Stewart and Cohe...	retrieval/ranking: no answer-containing passag...	increase corpus coverage, fetch object/related...
2	173697	Who was the composer of Into the Woods?	[Stephen Sondheim, Stephen Joshua Sondheim]	[On the Town (film), Chances Are (film), Imagi...	retrieval/ranking: no answer-containing passag...	increase corpus coverage, fetch object/related...
3	4657020	In what city was Jeanne de Casalis born?	[Basutoland]	[Anahí, The City (1916 film), Jamaica, London,...	retrieval/ranking: no answer-containing passag...	increase corpus coverage, fetch object/related...
4	764851	Who is the father of Peter II of Portugal?	[John IV of Portugal]	[Peter II of Portugal, Ptolemy II of Telmessos...	retrieval/ranking: no answer-containing passag...	increase corpus coverage, fetch object/related...

Next steps: Generate code with failures_df New interactive sheet

Case	Question	Failure Mode	Fix
0	Producer of The Pioneers?	Retrieval: corpus has no Franklyn Barrett page	Fetch object-entity Wikipedia page as additional corpus docu
1	Author of Revolution?	Retrieval: Russell Brand page absent from corpus	Expand corpus to include object-entity pages and top-N relate
2	Composer of Into the Woods?	Retrieval: Sondheim page missing; film page retrieved instead	Distinguish film vs. person entities during corpus construction
3	City where Jeanne de Casalis born?	Corpus gap: Basutoland/Lesotho not linked from her entity	Use full Wikipedia article (not just summary) for multi-hop fact
4	Father of Peter II of Portugal?	Ranking: correct passage present but ranked too low	Increase CANDIDATE_K or use a stronger cross-encoder mo

Recurring failure modes: (1) *Corpus coverage gaps* — the most common cause; Wikipedia summaries for obscure or long-tail entities are absent. (2) *Object-entity mismatch* — queries ask about a related entity not directly indexed. (3) *Ranking failures* — correct passages exist in the candidate set but are scored below the cut-off. All three modes would be partially addressed by using a full Wikipedia dump and a larger CANDIDATE_K.

Part 4 — Self-Reflective RAG and Final Analysis

Task 4.1 — Self-Reflective RAG

A self-reflective component was implemented as a second LLM call (REFLECTION_PROMPT) that critiques the draft answer on three axes:

- **Grounding:** every factual claim must be traceable to a cited [P#] passage; unsupported claims are removed.
- **Citation format:** at least one [P#] tag must be present; if absent, the model adds it or abstains.
- **Completeness:** the answer must address the question; if not, the model revises using the same passages.

The reflection stage never triggers additional retrieval — it operates solely on the top-5 reranked passages already retrieved. If after all corrections no passage can support a confident answer, the model writes the fixed abstention string. This bounds latency to exactly two LLM calls per question and avoids unbounded retrieval loops.

Reflection prompt logic (condensed): The model is given the question, all passage text, and the draft answer. It is instructed to check grounding, citations, and completeness in sequence, then return only the final corrected answer. Answers that pass all checks are returned unchanged; answers that fail are rewritten or replaced with the abstention string.

Before/after example: For "Who was the screenwriter for On the Town?" the draft answer "The answer is Betty Comden [P1]." passes all three checks and is returned unchanged. For cases where the draft lacked a citation (deterministic fallback path), the reflection stage replaced the answer with "Insufficient evidence in the retrieved passages." — correctly abstaining rather than allowing an unverifiable claim.

Task 4.2 — Comparative Evaluation

Figure 4.2 — Final 5-system comparison table across all retrieval configurations

	Recall@1	Precision@1	Recall@3	Precision@3	Recall@5	Precision@5	MRR
Dense baseline	0.27	0.27	0.30	0.116667	0.34	0.102	0.295000
Dense + query expansion	0.28	0.28	0.32	0.133333	0.34	0.100	0.303333
Hybrid RRF	0.26	0.26	0.33	0.130000	0.35	0.102	0.296167
Hybrid + reranker	0.32	0.32	0.35	0.146667	0.36	0.112	0.334167
Final system: hybrid + reranker + reflection	0.32	0.32	0.35	0.146667	0.36	0.112	0.334167

Axis	Observation
Quality	Hybrid + reranker achieves best retrieval metrics (Recall@1 0.32, MRR 0.334). Self-reflection adds generation-level improvements
Latency	Dense baseline is fastest (~1 embed call). Each stage adds overhead: BM25 scoring, cross-encoder inference, and a second LLM
Complexity	The final pipeline has 5 components (embedder, BM25, RRF, cross-encoder, LLM×2). Each is a potential failure point and requires
Cost	BM25 and cross-encoder run locally at zero API cost. Two LLM calls per question doubles generation cost vs. single-pass grounde

Task 4.3 — Final Discussion

Strengths. The pipeline layers complementary retrieval signals: dense embeddings capture semantic paraphrase similarity, BM25 anchors on exact entity tokens (critical for PopQA's proper-noun-heavy questions), and the cross-encoder reranks with full bidirectional attention. The self-reflection stage acts as a post-hoc grounding verifier, reducing citation-free answers without widening the retrieval pool. Metadata-driven query expansion leverages Wikidata aliases and property keywords that are already present in the dataset, requiring no additional model calls.

Limitations. (1) *Corpus coverage:* Wikipedia API summaries may omit the specific sentence containing the answer, capping Recall@5 around 0.36 even with the best pipeline. (2) *Answer matching:* Relevance is judged by normalised substring presence, missing paraphrases (e.g., "U.S." vs "United States") and multi-hop facts. (3) *Evaluation scale:* 100 questions may not fully represent the long-tail distribution; low-popularity entities are disproportionately hard.

Future improvement. Replacing the Wikipedia summary corpus with a full-article chunked Wikipedia dump (e.g., the standard 100-word chunks from KILT) would substantially increase corpus coverage and likely push Recall@5 above 0.60, closing the largest remaining performance gap. Adding entity linking (mapping mentions to Wikidata QIDs) would further improve query expansion and reduce the object-entity retrieval failures identified in the error analysis.